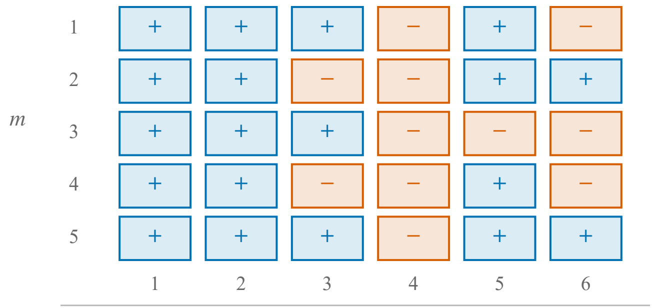
Reference-calibrated verification of metamodel Jacobians

Rank components after training; spend reference simulations only where calibrated evidence is insufficient.

a Cross-retrain evidence

Autodiff components from M independently trained models k



Candidate component scores

mean-aligned sign a_k

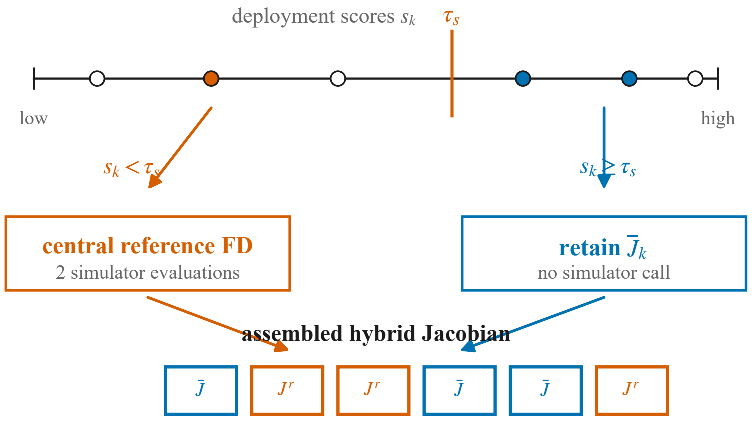
signal-to-noise $u_k = |\bar{J}_k|/\hat{\sigma}_k$

magnitude $m_k = |\bar{J}_k|$

b Calibrate, then deploy

Reference-labelled calibration queries

Choose score and τ_s for the intended loss; abstain if the risk screen fails.



c Measured operating point ($a_k \geq 0.9$)

benchmark	coverage	risk	evals/query
TMM near-saturated	95.9%	1.5%	0.90/22
TMM stressed	74.0%	13.4%	5.72/22
Heat/Poisson stressed	79.9%	4.9%	6.44/32

Score choice depends on the loss

ROC AUC	SNR: both TMM; magnitude: heat
fixed-budget error	sign: lowest in all three

$R_{FT} = P(\text{sign wrong} \mid \text{accepted})$

cost column: selective / complete central FD

Scope boundary common positive rescaling of all members at fixed k $\Rightarrow$ unchanged sign agreement and SNR; m_k changes

shared bias $\Rightarrow$ high agreement can still be wrong

Evidence screen: 3,580/3,580 external labels passed step/grid checks; unresolved antenna derivatives were excluded.

calibrated triage $\neq$ correctness certificate